



Society of Petroleum Engineers

SPE-226915-MS

Characterizing the Tinat Member: An Interdisciplinary Study Combining Sedimentology, Stratigraphy, and Petrophysics

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/226915-MS](https://doi.org/10.2118/226915-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Interacting fluvial, aeolian, and marine sedimentation processes typically yield complex, heterogeneous depositional sequences characterized by lateral and vertical variability. Consequently, precise stratigraphic correlations assume critical importance in predicting rock quality and connectivity, thus better characterization. However, correlating surfaces within deposits of terrestrial origin poses significant challenges. To address that, this study introduces a systematic approach on defining key stratigraphic bounding surfaces through integrating relationships among facies stacking patterns, correlative surfaces, and subsurface data.

For the purpose of this study, more than 1000 ft. of core data were examined. Facies associations and major stratigraphic markers were defined, honoring regional sedimentology and stratigraphic understanding. Stratigraphic markers were then used as correlative surfaces to build frameworks across the field with the integration of wireline logs responses, facies stacking patterns, complemented by formation pressure data as validation tools for correlation and sub-unit divisions. Quality evaluation through the utilization of conventional core analysis and petrographic analysis also added to the study to have a comprehensive understanding about heterogeneity in relation with different stratigraphic sub-units and facies associations.

Four surfaces were identified, including Major Surface 1, which corresponds to a 1st-order sequence, and three stratigraphic sub-unit surfaces, representing 4th-order sequences, derived from core. The remaining major surfaces were identified based on log characteristics, supplemented by regional geological understanding. The upper boundary of Tinat Member is marked by a pebbly conglomeratic unconformity (1st-order sequence), marking the transition between AP5 and AP6, as defined by Arabian Plate tectonostratigraphic nomenclature. Three additional surfaces were recognized from core data: Sub-Unit 1, distinguished by an abrupt textural shift coinciding with the presence of paleosols; Sub-Unit 2, identified as a regional wetting phase, characterized by the deposition of playa facies, likely linked to a rising water table; and Sub-Unit 3, representing a sudden transition from predominantly aeolian-marine setting to aeolian settings. The lower boundary of the Tinat Member is bounded by an unconformity recognized from high gamma-ray (GR) log responses corresponding to red siltstone deposits of interfluvial origins. The dominant depositional environment is continental, although evidence suggests intermittent aeolian-marginal marine

interactions, manifesting as wetter aeolian system with increased muddy facies, bioturbation, and tidally influenced deposits. Pressure data verifies the subdivision of these three units, indicating variations in depositional processes which contribute to understand compartmentalization.

The Tinat Member exhibits a distinct progradational stacking pattern, transitioning upwards from an aeolian-marginal marine system to a playa, and aeolian setting and finally terminating in interacted fluvial-aeolian system. Notably, the uppermost part of the Tinat Member is capped by a paleosol section, indicating a period of non-deposition preceding the deposition of the AP6 Megasequence. Three identifiable stratigraphic sub-units can be correlated, allowing for a more accurate assessment and subsurface modeling.

Introduction

Rock behavior and performance analysis can be complex, but sedimentology can play a crucial role in predicting quality and connectivity. By integrating sedimentology with petrophysical data, petrography analysis, and formation pressure data, a more comprehensive understanding of the subsurface data can be achieved. Sedimentology provides a broad understanding of depositional environments, facies distribution, and architectural elements that control rock quality and connectivity. In contrast, petrophysics and petrography analysis offer detailed information on the physical properties of rocks, including porosity, permeability, and mineral composition. The integration of these disciplines, combined with formation pressure data, enables the identification of potential flow barriers and optimization of production strategies. This study aims to identify key factors for dealing with heterogeneous settings deposited in continental arid to semi-arid environments.

Sedimentology and stratigraphy are crucial for defining the depositional environment, facies associations, and stratigraphic markers, which form the basis of subsurface characterization (e.g., [Al-Masrahy, et al., 2024](#)). The primary objective is to identify correlative surfaces within continental deposits, which are currently absent, and subsequently divide the member into sub-units, propagating them vertically and laterally to non-cored wells based on stacking patterns and log responses. To support geological interpretation, this study integrates formation pressure data and rock properties derived from conventional core analysis (CCA) and petrography data to validate the stratigraphic subdivision.

The interaction of aeolian, fluvial, and marine processes significantly influences reservoir characterization by creating a complex architecture with varied rock properties ([Jordan and Montney, 2010](#); [Rodriguez-Lopez, et al., 2012, 2013](#); [Al-Masrahy and Mountney, 2015 a,b](#); [Al-Masrahy, 2017](#)). Aeolian deposits, formed by wind, often consist of well-sorted sands with high permeability, while fluvial deposits from rivers may have more varied grain sizes and lower permeability. Marine environments contribute deposits that can act as barriers or seals, especially when cemented by minerals. In transitional zones, these environments overlap, resulting in diverse facies, including fluvial channels, marine muds, and aeolian dunes. These transitions can lead to intricate geometries and spatial variations in porosity and permeability, affecting fluid flow dynamics ([Figure 1](#)).

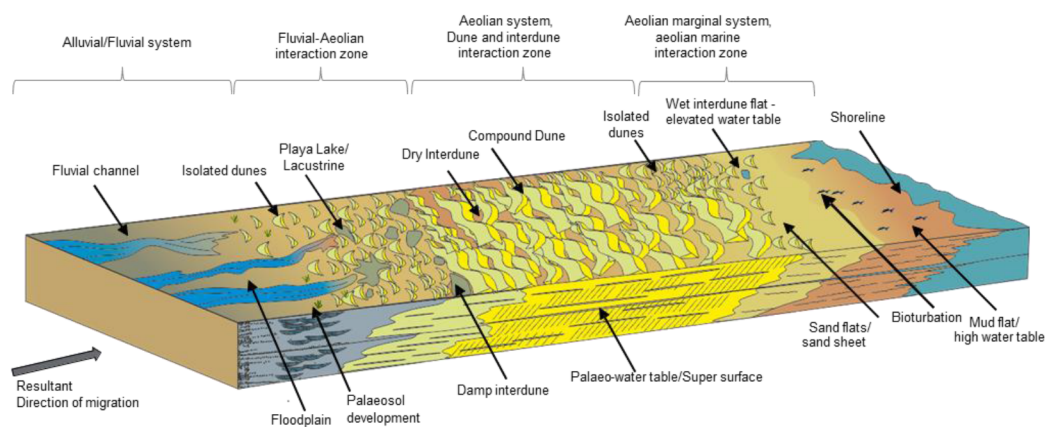


Figure 1—A schematic model illustrates the system interactions between aeolian, fluvial, and marine depositional settings, as well as the expected resultant subsurface expressions (From, Al-Masrahy, 2020).

Geological Background

The Arabian Plate experienced a significant tectonic event during the Mid-Carboniferous period, triggered by the Hercynian Orogeny. This event led to substantial uplift and erosion, followed by continental deposition (Konert et al., 2001; Al-Husseini, 2004). Sharland et al. (2001) identified several Tectonostratigraphic Megasequences (TMS) in the Arabian Plate, including TMS AP5, which encompasses the Permian-Carboniferous sand-rich interval. This megasequence is bounded by the Hercynian Unconformity at its base and the Pre-Khuff Unconformity at its top, marking the boundary between TMS AP5 and TMS AP6, and representing a first-order cycle. The Permian-Carboniferous interval is further subdivided into two formations: the Juwayl and Nuayyim Formations. The Nuayyim Formation is itself divided into two members: The Late Sakmarian Wudayhi Member and the Artinskian to Kungurian Tinat Member (Figure 2).

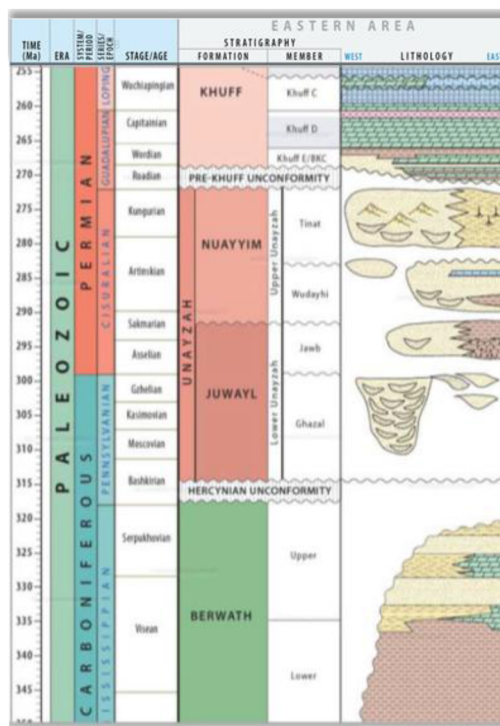


Figure 2—The late Carboniferous – Permian general stratigraphy (Saudi Aramco 2016).

The Tinat Member time period is characterized by the extensive development of aeolian dune fields in sub-tropical to semi-arid regions, specifically between 15-35°S paleolatitudes (Konert et al., 2001). However, evidence from core samples suggests that the lower Nuayyim Formation experienced periodic, likely brief, marine incursions, as indicated by bioturbated horizons (Polo et al., 2018, 2019). Although most of the interaction occurred as a result of fluctuations in the water table, which controlled the system (Al-Masrahy, 2017, 2020). This depositional environment is similar to those found on the Arabian Plate today, which is currently located at approximately 30°N (Al-Masrahy, 2012, 2020). The vast dune fields, interspersed with ephemeral fluvial streams and playa lakes, may have had marine interactions towards the east and south of the Arabian platform. The system can be compared to the modern-day dunes, playas, and sabkhas of the northern Rub' al-Khali desert system (Al-Masrahy and Mountney, 2013).

The depositional environment of the Tinat Member was characterized by arid to semi-arid continental settings, featuring ephemeral lakes, ephemeral fluvial, and a variety of aeolian deposits (Melvin et al., 2010; Al-Masrahy, 2012, 2020). Tinat Member comprises various aeolian facies associations from erg-center to erg-margin deposits, such as dunes, interdunes, sand-sheets, and marginal flats deposits influenced by aeolian-marine interactions, resulting in a complex and dynamic depositional environment.

Methodology

This study utilized an inter-disciplinary approach combining sedimentology, stratigraphic, petrophysical analysis, and dynamic data to provide a comprehensive understanding of the Tinat member's complex architecture (Figure, 3). The investigation began with a detailed examination of over 1000 feet of cores from five wells, focusing on the fundamental sedimentological components of depositional environment, facies associations, and stratigraphic units or correlative surfaces. Building on this foundation, wireline log analysis and core plug data were used to characterize facies properties and identify vertical and lateral stacking patterns. The analysis was further enhanced by incorporating subsurface data, allowing for a thorough understanding of rock heterogeneity in relation to different stratigraphic sub-units and facies associations.

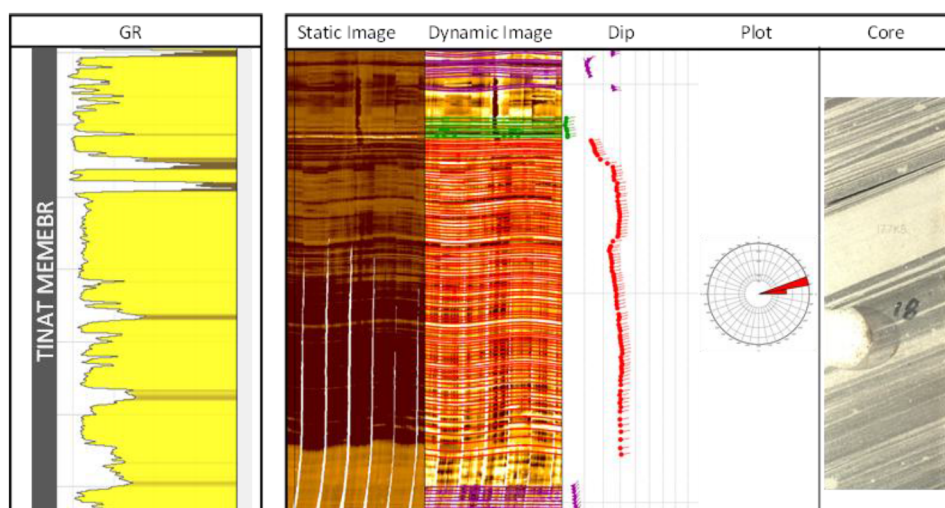


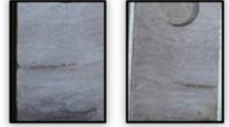
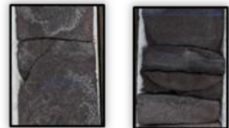
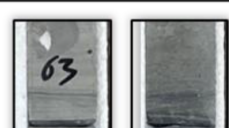
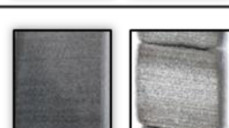
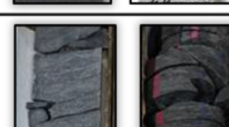
Figure 3—The core data has been calibrated to match the Gamma Ray (GR) log and borehole image logs, ensuring accurate and reliable information.

Results & Discussion

Facies Associations

Building on previous work and regional geological context, sedimentological core data observed a strong continental deposition processes that occurred during Tinat Member period and yielded in 9 distinct facies associations (Table 1):

Table 1—Facies associations

FACIES ASSOCIATION	CODE	LITHOFACIES	DESCRIPTION	CORE IMAGE
Paleosols	F1	Pedogenic altered argillaceous sandstone	Reddish brown, poorly sorted and deformed textures	
Fluvial Channel	F2	Trough cross-bedded and scoured base sandstone	fine -medium grained sandstones. Upward fining succession with rip-up clasts and scoured base, trough cross-bedding sets	
Flood Plain	F3	Irregular laminated sandstone	Siltstone - very fine grained sandstones, irregular laminated, fractured, intercalated with upward fining succession of trough cross-bedded sandstones	
Aeolian Dune	F4	Cross bedded sandstone with soft sediment deformation in part	Fine - medium grained sandstones. Forming cross-bedded sets alternating with irregular - horizontal laminated sandstones.	
Interdune	F5	Irregular – flat laminated sandstone	Very fine - fine grained sandstone with irregular/crinkly - horizontal laminated sandstones in between dunes	
Sand Sheet	F6	Flat to irregular laminated sandstone	Fine - medium grained sandstones, irregular - flat laminated.	
Playa	F7	Irregular – undulated to massive siltstone	Siltstone, irregular - wavy and horizontal lamination	
Tidal Flats	F8	Irregular – flat laminated bioturbated sandstone, and Massive – thinly flat laminated siltstone	Fine grained, flat - irregular laminated sandstones with bioturbation? and mud drapes?, massive - flat laminated, intercalated within irregular laminated siltstone.	
Tidal Channel	F9	Cross bedded with soft sediment deformation in part and scoured base sandstone	Fine – medium grain sandstone, upward fining succession with hint of regularly spacing lamination and bioturbation.	

Paleosols (F1). Paleosols showed reddish brown to gray color, very fine to coarse grain argillaceous sandstone to occasionally mudstone, pedogenic altered with rootlet traces, cracks, slickensides, and mainly stacked variegated – mottled structures.

This facies association is not necessarily depositional facies but required to be distinguished due to its significance in the evaluation of the rock quality and stratigraphic application. Paleosols formed because of the physical, biological, and chemical modification of sediment or rock exposed at the earth surface (Wright, 1992). Frequently found at the upper most interval with no aeolian facies above, the presence of paleosols were associated with the regional unconformity and marked the top of Tinat member prior to deposition of Khuff formation and can be correlated regionally although vary in thickness.

Fluvial Channel (F2). Light brown, fine to medium grain, apparent stacked upward fining succession with trough cross-beds, rip-up clasts, and erosional base. Multiple scouring and soft sediment deformation also recorded. These deposited facies are attributed to ephemeral fluvial system (cf. Grace et al., 2025).

Fluvial channel facies association exhibited upward fining sequences consists of alternating high energy channel-fill observed from erosional base, lag deposit, and trough cross-beds alternating into low-energy into overbank deposits with current ripple and mud drapes at the top. This fluvial channel associated with flood plain indicated an established river and appeared in only in the upper section of Tinat member.

Flood Plain (F3). Dark brown, silt to very fine grain with irregular laminations and fractures. Flood plain associated with upward fining succession of fluvial channel sandstones and it appeared as thin layers at the very top of the interval and marked the abandonment at the end of channel-fill cycle. It forms when river overflows its banks and deposits fine sediment onto flat or gently sloping of surrounding area (cf. Lopez, and Mader, 1985; Grace et al., 2025),

Aeolian Dune (F4). Light yellowish grey of fine to medium grain sandstone, homogenous well rounded and very well-sorted texture with tabular cross-bedded sets indicating multiple grain size segregation as well as closely spaced millimeters thick laminations or pin stripe lamination (Fryberger and Schenk, 1988) and inversely graded, occasional soft sediment deformations are also observed.

Thick accumulations of fine to medium grain high angle tabular cross-bedded sets are interpreted as aeolian dune deposits (Mountney, 2006), the very well sorted sands with alternating thick and thin sand beds interpreted as grain flow cross strata produced at the dune fore-sets (cf Al-Masrahy, 2020; Grace et al., 2025). In the study area, aeolian dunes can be subdivided into two depositional settings i.e., the inland proximal dune and distal marginal dune-field where aeolian processes invaded by fluctuated water table affected by tides (cf. Fryberger et al., 1990b), the latter is similar to aeolian dune described previously, only differ in appearance of thin inclined silt drapes, regularly spacing laminations that show patterns of increasing and decreasing fore-sets (rhythmicity), and exists only in the lower most portion of Tinat member.

Interdune (F5). Light grey, very fine to fine grain sandstone with irregular to flat laminated and occur interbedded with aeolian dune as thin layers.

Fine grains with irregular to flat laminated suggested a low energy deposition occurs in the depression or low-lying area associated with aeolian dunes complex. This accumulation can be located in the topographically low area affected by water table fluctuation and water table fringe (Mountney 2012, Al-Masrahy and Mountney, 2015) that reflected in mud to silt size sediment and laminations.

Sand Sheet (F6). Gray sandstone, medium grain size, moderately to well sorted, flat laminated to sub-horizontal laminated, wind ripple laminations.

Sand sheet is considered as deflation or bypass zone in aeolian deposit and often marginal facies to dune complexes and showing gently inclined ripple or flat lying strata (Fryberger, 1990a). Sedimentary features are similar to dry interdune. However, sand sheets are laterally extensive and commonly thicker and even coarser than interdune deposits. Sand sheets are sandy plains whose topographic expression is independent

of water table and tend to develop from remnant deposits of migrating dunes (cf. Fryberger et al., 1983; Basilici, 2009).

Playa Lake (F7). Black to dark gray siltstone, irregular laminated to massive, occasional streaks of very fine to fine sand laminated, and sharp contact with other facies associations.

The siltstone dominated interval represents the semi-arid ephemeral lake deposits, this playa facies has been interpreted to be direct reflections in the level of the paleo-water table throughout the time of its deposition and correlatable as internal stratigraphy within aeolian-dominated units (cf. Lopez, and Mader, 1985; Melvin et al., 2010).

Tidal Flats (F8). Consists of gray to dark gray sandstone, argillaceous laminated sandstone appeared as irregular black thin lines – current ripple – wavy laminations, bioturbation is sparse to uncommon (bioturbation index 1-2), and intercalated with dark gray siltstone, massive to irregular laminated, appeared as thin layers within tidal flat and exhibits a siltier portion of sand flat facies whenever there is a low energy and only suspension can be deposited.

Tidal flats are predominantly found in the lower portion of the Tinat Member. In this study, we distinguish between tidal flats and sand sheet facies to account for the unique sedimentary features that interpreted to occur in the extensive, low-lying flat areas of the aeolian and marginal marine system, where there is a prominent influence of water table that episodically or continuously invaded by tides (cf. Fryberger et al., 1990b, Al-Masrahy, 2017). Pervasive argillaceous laminations, current ripples to wavy laminations, and bioturbation suggested a strong water-lain sediment compared to sand sheet which interpreted above and geographically located more inland or in the proximal part of aeolian system, (cf. Rodriguez-Lopez et al., 2012). cf. Polo et al., 2020).

Tidal Channel (F9). Brown – light brown argillaceous sandstone, moderately sorted, cross bedded, soft sediment deformation, stacked upward fining succession with erosional base, tidal laminated rhythmites, and sparse bioturbation, (cf. Rodriguez-Lopez et al., 2011; Rodriguez-Lopez et al., 2012; Polo et al., 2020).

This facies association occurred only at the lower most portion of Tinat member, associated with aeolian dune and tidal flat that constitute aeolian system encroaching into marginal marine zone (cf. Fryberger et al., 1990b; Rodriguez-Loper et al., 2011).

Depositional Environment

A number of distinctive depositional facies were identified in previous section and those facies indicated an aeolian dominated process and deposition within an arid to semi-arid environment. Schematic diagrams illustrating subdivisions of major aeolian dune-field (erg) system have been proposed by Rodriguez-Lopez et al. 2012, Al-Masrahy 2020, as well as Cosgrove et al 2021, and many others. The main idea is to construct aeolian depositional model based on its relative proximity from the inland continental highland down sloping onto coastal margin or from back-erg, central erg, and marginal fore erg or coastal erg zone.

Vertical and lateral interaction of each facies associations suggested a depositional pattern subdivision that related to the proximal to distal relative position of the erg-system from further continental inland depositional system to marginal end system. Three subdivisions (Figure 6) identified from this study i.e. Fluvial – Aeolian system (back-erg), Aeolian system (central-erg), and Aeolian marginal zone (fore-erg/coastal).

Central Aeolian (Central-erg)

The central aeolian depositional complex is a region of high sand transport, where wind-blown sediment dominates the setting, forming a vast expanse of complex dune systems with high density (see [Al-Masrahy and Mountney, 2013](#)). Notably, the middle interval of the Tinat Member in the study area is characterized by the preservation and dominance of these aeolian dunes. These tabular-planar cross-bedded dune sands do not occur as isolated, single entities, but rather as multiple, stacked, and interconnected sand bodies that dominate the strata, and exhibit only minor interdune and sand flat deposits. This contrasts with dunes deposited at the marginal areas of the dune field, particularly in mixed fluvial-aeolian settings, where dune morphology tends to be more isolated and thinner ([Figure 1](#)).

Mixed Fluvial – Aeolian setting (Back-erg)

Fluvial – aeolian depositional zone serves as the proximal part of the aeolian dune field system. This dynamic relationship where rivers entering dunes complex resulting in the formation of distinctive facies associations. Wind-blown deposits tend to be relatively well-sorted compared to sediment deposited by other processes, such as fluvial activity ([Masrahy, 2017](#)). Generalized schematic model illustrated depositional process taking place at the dune-field margins illustrated by [Al-Masrahy and Mountney 2015](#), shows more isolated dunes and more developed marginal facies deposited in the low-lying flat area such as playa lake, flood plain, sand sheet with occasional fluvial channel, which is coherent with the facies association found in the study area. Sand sheet acted as dominated background facies and alternating with aeolian dune-interdune facies and occasional river stream deposits of fluvial channel and flood plain. These facies interactions suggest a back dune-field margin depositional environment (back-erg) and can be found at the upper interval of Tinat member, ([Figure 4](#)).

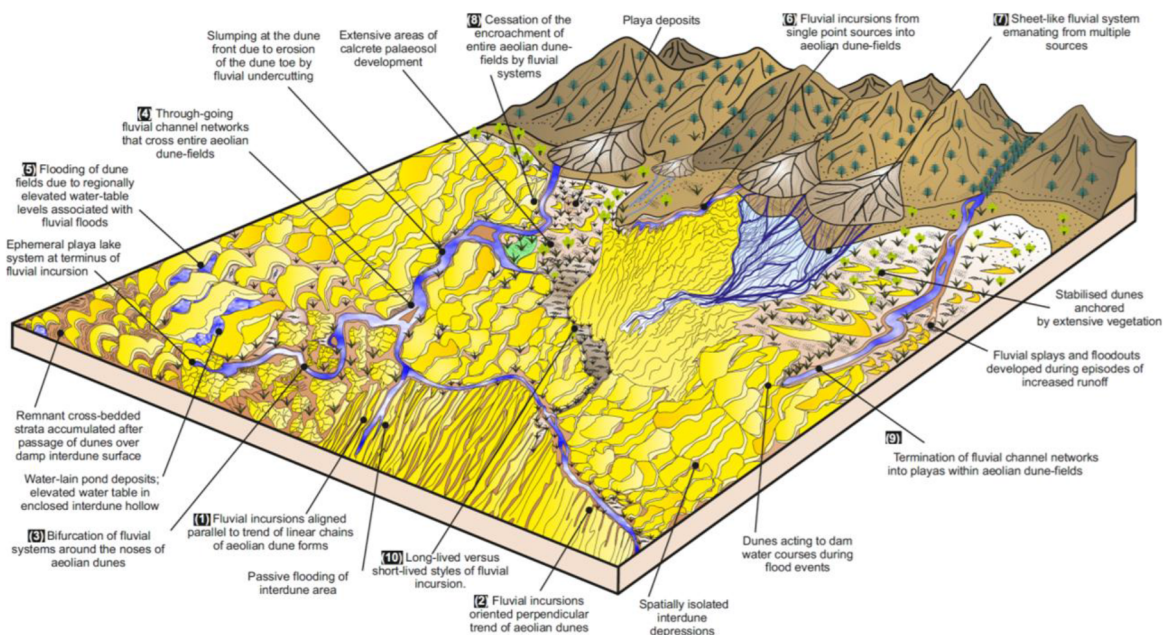


Figure 4—Schematic model illustrating different types of aeolian-fluvial system interaction. Numbers in black boxes relate to the ten types of fluvial-aeolian system interaction discussed in [Al-Masrahy and Mountney, 2015](#), and [Al-Masrahy 2017](#). (Figure modified from [Al-Masrahy and Mountney, 2015](#)).

Aeolian – Marine Interaction setting (Fore-erg)

Previous work by [Polo, 2020](#) identified aeolian-field margin and shallow marine interaction based on sedimentology and ichnology in Tinat member, [Fryberger et al 1983](#) studied modern sedimentary facies of complex offshore prograding aeolian sand sea along portions of the Arabian Gulf coastline, East Saudi Arabia, and meanwhile some study had been done by several authors regarding marine incursion over aeolian erg margin systems whether in ancient or recent examples. [Al-Masrahy \(2020\)](#) presented a modern analogue that illustrates the impact of water table fluctuations on the distribution of aeolian deposits in a direction towards the marine shoreline ([Figure, 1](#)). This analogue demonstrates how changes in the water table can lead to changes in facies and architectures ([Figure 5](#)). Yet the understanding of aeolian – shallow marine interaction still not as advance as compared to inland continental depositional environment ([Rodriguez-Lopez, et. al., 2012, 2013](#)). The interaction between aeolian sedimentation and tidally influenced shallow marine deposits identified from well cores data. The stratification types commonly found in aeolian sedimentation such as high angle cross-bed formed by grain flow in the lee face as well as wind-ripple lamination showing inverse grading can be identified in this interval and conformably alternating with some interval of bioturbations, inclined thin carbonaceous laminations, double silt drapes, and patterns of regularly spacing of thin and thick lamination sets suggesting tidally influence subaqueous deposition.

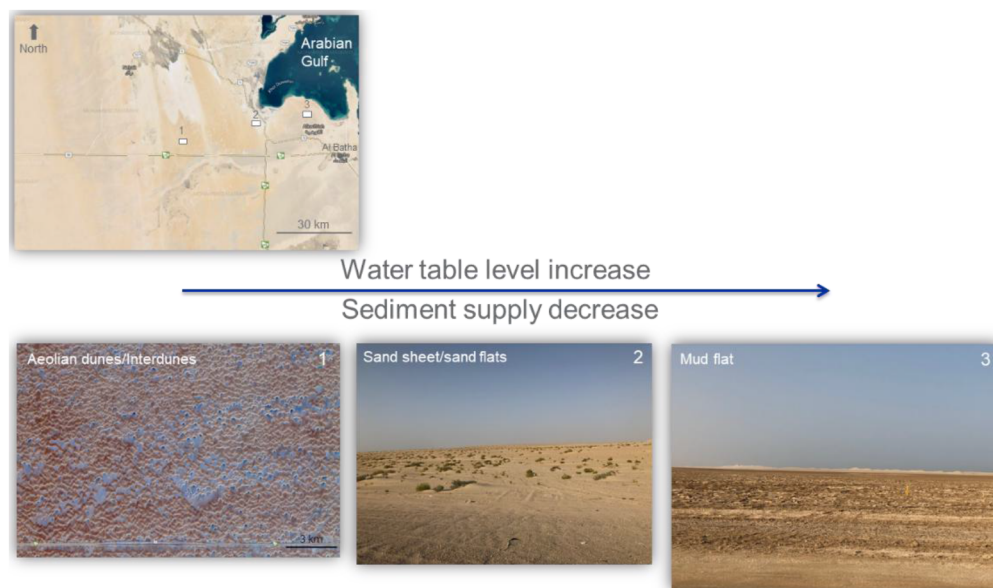


Figure 5—Modern analogues demonstrate the lateral transition of the aeolian system, highlighting the impact of the water table on the distribution of aeolian deposits from central to marginal areas, where they interact with the marine system and an elevated subsurface water table (Figure from [Al-Masrahy, 2020](#)).

As the water table rises or falls, it influences the distribution of aeolian deposits, leading to a variety of facies and architectures. By studying these modern analogues, we can gain a better understanding of the ancient systems and improve our predictions of subsurface counter parts.

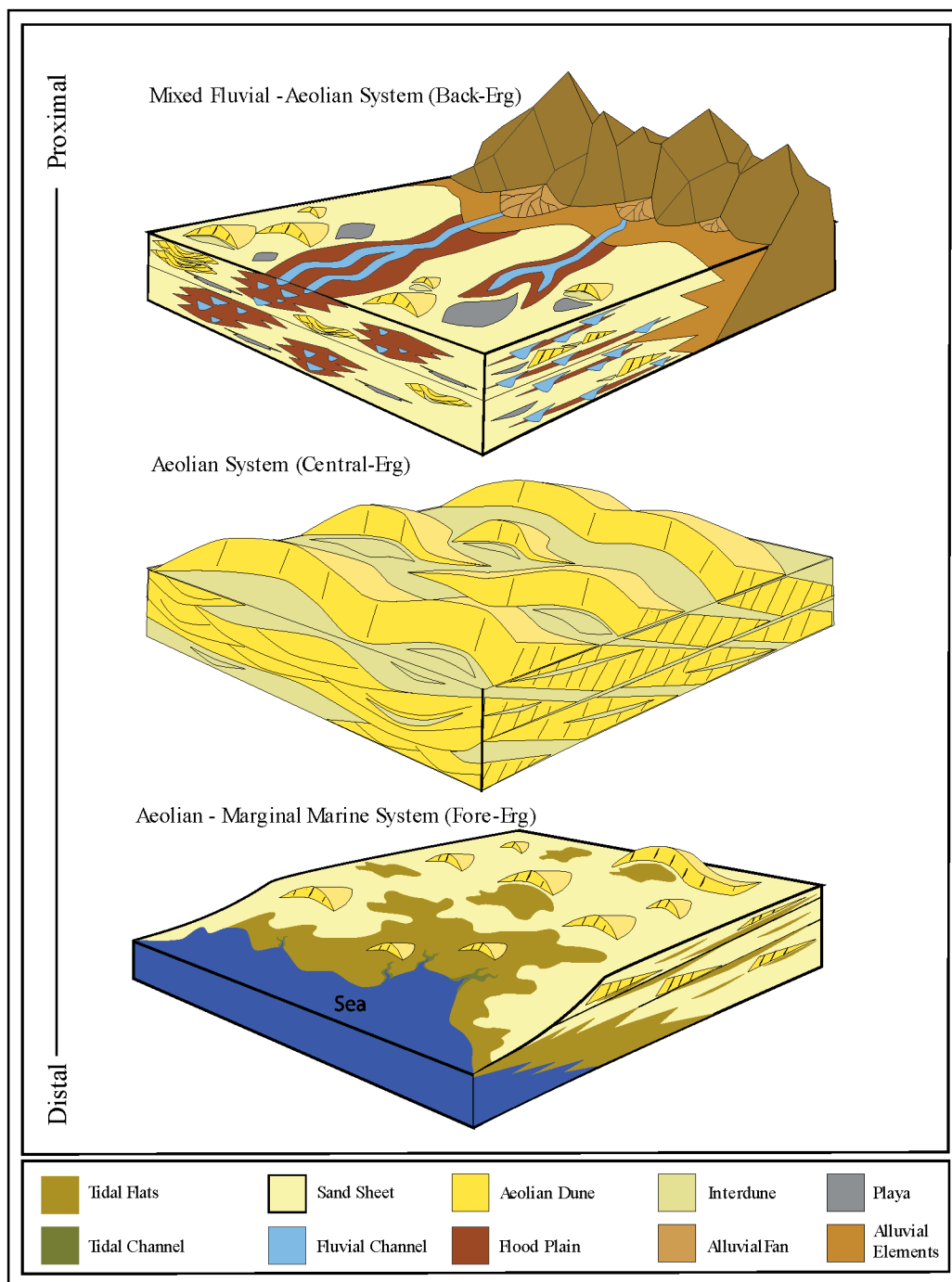


Figure 6—Schematic diagram illustrating sub-divisions of aeolian dune-field (erg) systems in study area. Not to scale, (Proximal and distal see Figure 1).

Stratigraphic Sub-Divisions and Facies Stacking Patterns

Facies associations interpreted from well cores are organized in several stacked cycles and bounding surfaces. Kocurek (1996) defined four types of bounding surfaces in dune sands: reactivation surfaces, superposition surfaces, interdune surfaces, and super surfaces. Clemensen and Tirsgaard (1990) signified sand-drift surfaces to separate aeolian strata unit. Melvin (2010), Polo (2020), and Rodriguez-Lopez (2012) offered a prolonged wet-substrate deposits such as wet-interdune and or playa lake within aeolian dominated deposits as the continental expression of marine transgression and reflected in fluctuations in the paleo-water table.

Four surfaces were identified from well cores and consisted of one major surface which corresponds to 1st – order sequence and marked the top boundary of AP5 TMS, and three aeolian stratigraphy sub-units or bounding surfaces representing 4th order sequence.

Started from the bottom of Tinat member, Sub-Unit 3 (SU3) represents the first erosional surface (Figure 7, A) attributed to marine related water table interaction in marginal dune-field areas (fore-erg) to aeolian dominated setting in central dune-field (central-erg) by the first occurrence of aeolian dunes and sand sheets overlying muddy interval of possibly tidal sand flats. Shifting in gamma-ray logs baseline marking this transition of different genetic unit.

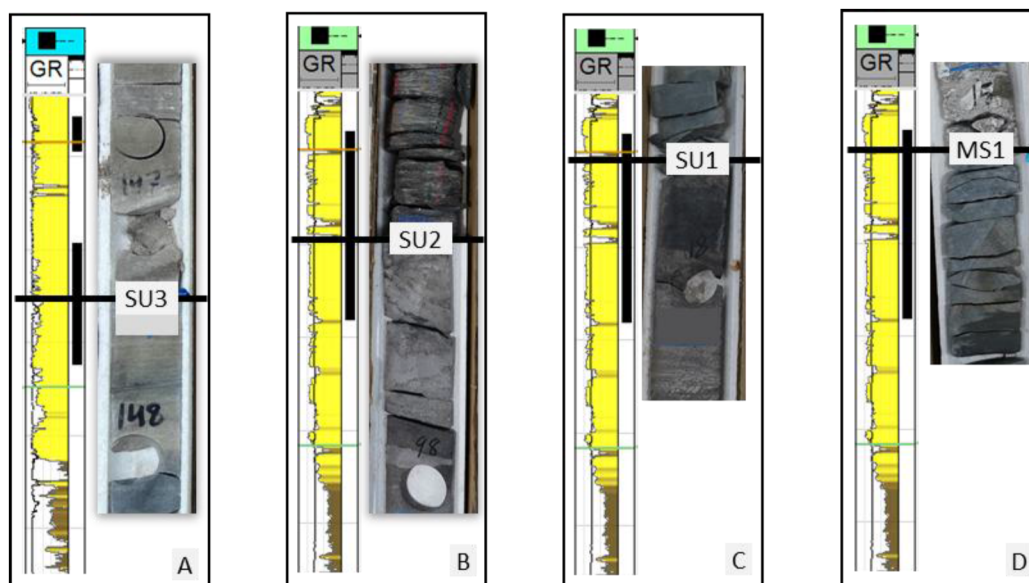


Figure 7—(A) SU3 bounding surface. (B) SU2 bounding surface. (C) SU1 bounding surface. (D) MS1 3rd order sequence surface.

In the middle of Tinat member, Top of Sub-Unit 2 (SU2) (Figure 7) defined as the central aeolian dune-field (central-erg) and characterized by multiple stacked dunes bodies and bounded at the top by playa facies. This playa facies linked to rising paleo-water table, likely defined as inland continental expression of marine transgression in further shoreline and can be identified regionally nearby the study area from its distinctly high GR signature.

Top of Sub-Unit 1 (SU1) recognized by abrupt textural shift coinciding with the beginning paleosols development at the upper part of Tinat member (Figure 7, and 8). Although, the bottom of fossil-soil horizons is not necessarily served as chronostratigraphic timeline, since the vertical effect of pedogenic process or soil formation might different from one place to other, but this section needs to be distinguished due to its significance impact. SU1 consisted of mixed fluvial-aeolian deposits and marked the geomorphic changing from the central dune-field (central erg) in SU2 to proximal dune-field margin (back erg). Evidence of fluvial process recognized from multiple stacked of upward fining succession constitutes channel-fill and flood plain deposit entering the aeolian dominated environment composed of sand sheets and dune-interdune facies.

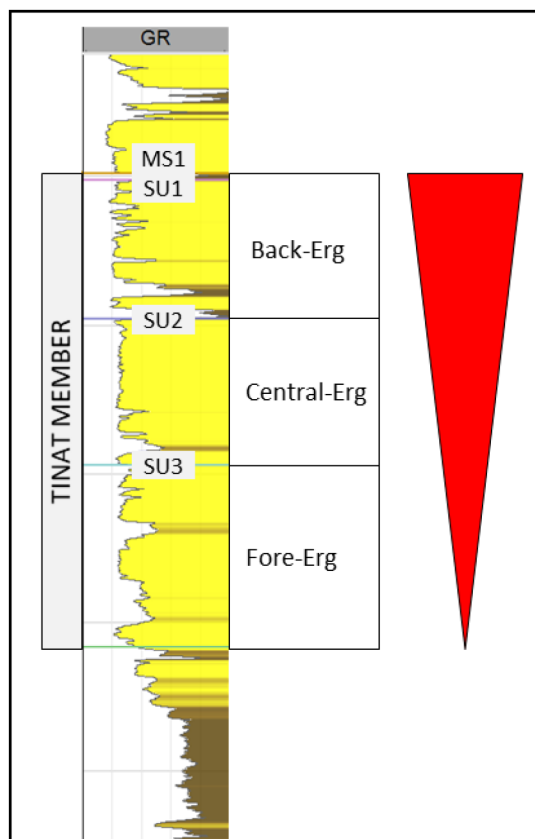


Figure 8—Stratigraphic sub-divisions and stacking patterns.

Above SU1 exhibited thick and pervasive paleosols section seen in many places at the top of Tinat Member. Melvin (2010) and Al-Masrahy et al., (2012) documented the end of deposition in Tinat member as a prolonged period of minimal deposition to non-deposition manifested in wide spread extent of paleosols development which indicated decrease of aridity in the area. Polo (2019) suggested that widespread paleosols on the top of Tinat Member presented as palimpsest surfaces wherein biogenic modifications demarcate the end of the whole sedimentation cycle. Paleosols at the top of Tinat Member sedimentation cycle marked the boundary of TMS AP5 and it is interpreted to be the evidence of a major unconformity associated with the continental rifting and spreading which culminated in the regional Pre-Khuff unconformity (Sharland et al., 2001). The major unconformity at the very top of paleosols defined as Major Surface 1 (MS1) (Figure 7 and 8) and characterized by pebbly conglomeratic sandstone marking the beginning of Khuff Formation.

Formation Evaluation and stratigraphic compartmentalization

Identifying flow unit within a subsurface rock unit involves a comprehensive analysis that combines stratigraphy, sedimentology, structural geology, petrography, petrophysics, and field dynamic performance (Ebanks et al., 1992). Formation pressure data acquired from formation tester tool were utilized to confirm the sub-divisions of Tinat Member. Initial and after production pressure data were used to compare and figure out different compartmentalization and its relationship with stratigraphic sub-divisions mentioned previously. There is a strong relationship between stratigraphic sub-divisions and hydrodynamic units (Figure 9), in particularly after production data that validates Tinat Member vertical separation into SU1, SU2, and SU3 as well as the section above MS1.

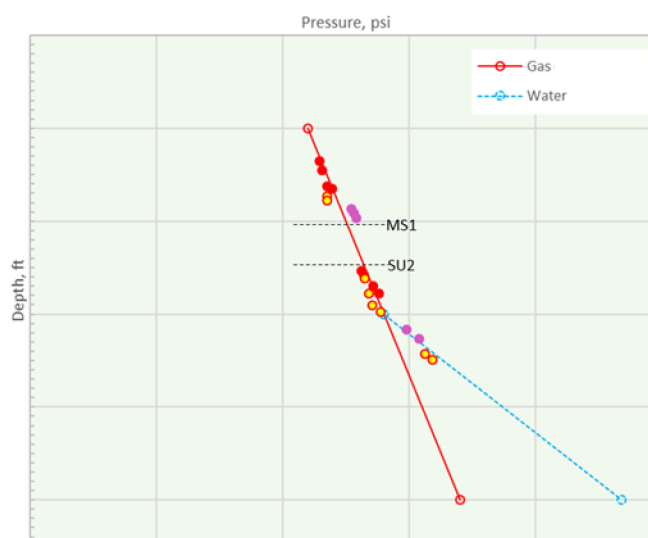


Figure 9—Initial formation pressure plot demonstrating the stratigraphic sub-divisions.

Subsurface Implications

Combining conventional core analysis (CCA) and petrography analysis provides a more comprehensive understanding of rock quality. By calibrating porosity and permeability data from core plugs with petrographic analysis, a detailed evaluation of the parameters impacting the quality can be conducted. This integrated approach enables the assessment of key factors such as mineral composition, pore space arrangement, cementation, and clay type, which are crucial in determining the unit's storage and flow capacity.

Representative samples from each sub-unit were analyzed to confirm the proposed sub-units. For instance, petrographic analysis of samples from Sub-Unit 2 (SU2), which represents aeolian dune facies (F4), revealed well-preserved intergranular and dissolution porosity. Scanning Electron Microscopy (SEM) data showed well-developed illite coatings on grains, which inhibited quartz overgrowth. This suggests that the rock quality in SU2 is relatively good, with preserved porosity and permeability. The integration of CCA and petrography data provides a robust framework for evaluating subsurface depositional succession quality and understanding the factors that control hydrocarbon storage and flow behavior.

Conclusions

The Tinat Member is a complex and heterogeneous depositional sequence that requires a comprehensive understanding of its stratigraphy, sedimentology, and petrophysics. This study provides a systematic approach to characterizing the Tinat Member, integrating relationships among facies stacking patterns, correlative surfaces, and subsurface data. The results of this study have significant implications on subsurface characterization, highlighting the importance of considering the intricate interactions between aeolian, fluvial, and marine processes in understanding the depositional architecture and rock quality.

Key points:

- **Depositional Environment:** The Tinat Member was deposited in an arid to semi-arid continental setting, featuring ephemeral lakes, ephemeral fluvial systems, and a variety of aeolian deposits, and Nine distinct facies associations.
- **Stratigraphic Sub-Divisions:** Four surfaces were identified, including one major surface corresponding to a 1st-order sequence and three stratigraphic sub-units representing 4th-order sequences.

- Facies Stacking Patterns: The facies associations are organized in several stacked cycles and bounding surfaces, with the central aeolian dune-field (central-erg) characterized by multiple stacked dune bodies and bounded at the top by playa facies.
- The study confirms the proposed sub-units and provides a detailed evaluation of the parameters impacting rock quality, such as mineral composition, pore space arrangement, cementation, and clay type.
- Tinat Member can be subdivided based on its correlative bounding surface into three identifiable stratigraphic sub-units, enabling a more precise evaluation of depositional elements compartmentalization and providing valuable insights for reservoir characterization.

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